# SPCD fMRI results interim summary

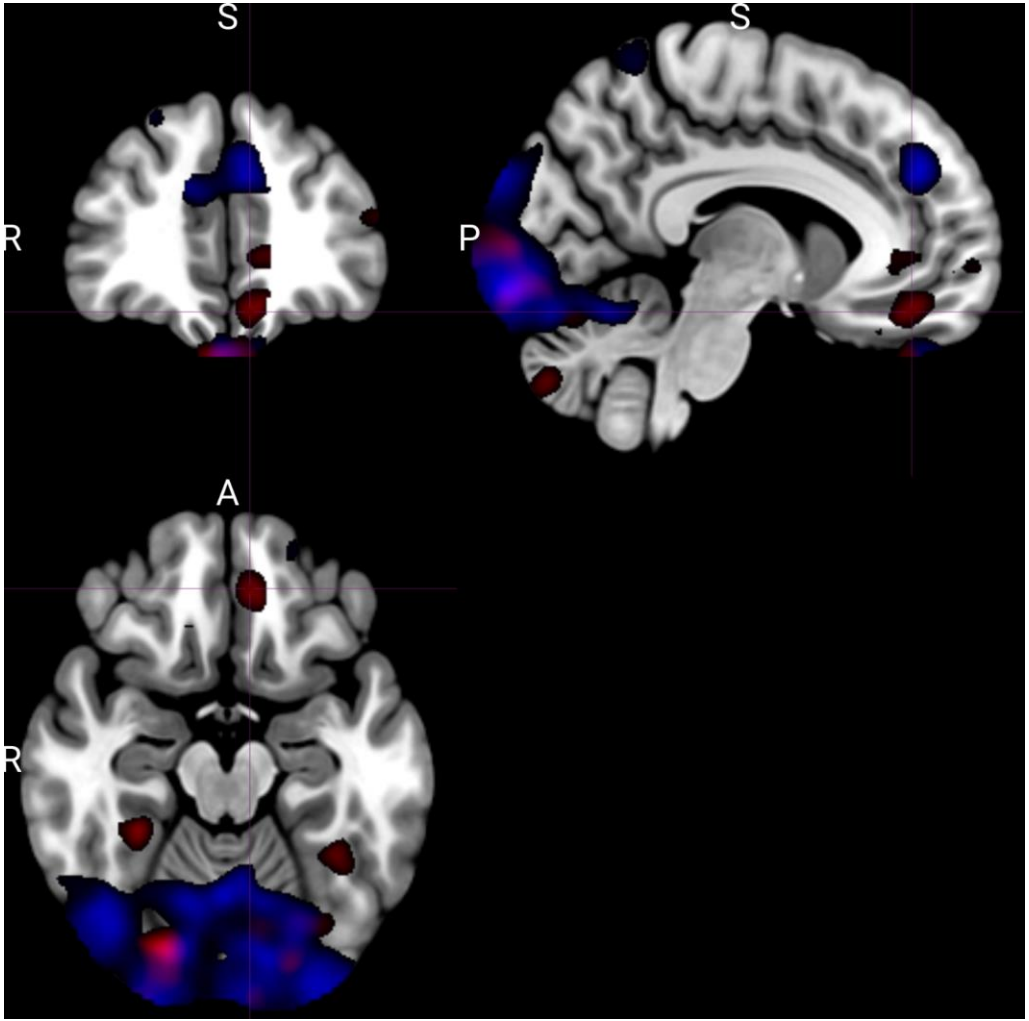
Qingfang Liu
June 11, 2026

Given that we found preconditioned cues encode reward identity according to behavioral results, where is the reward identity of preconditioned cues encoded? Is that encoded in the same region as the conditioned cues or different region?

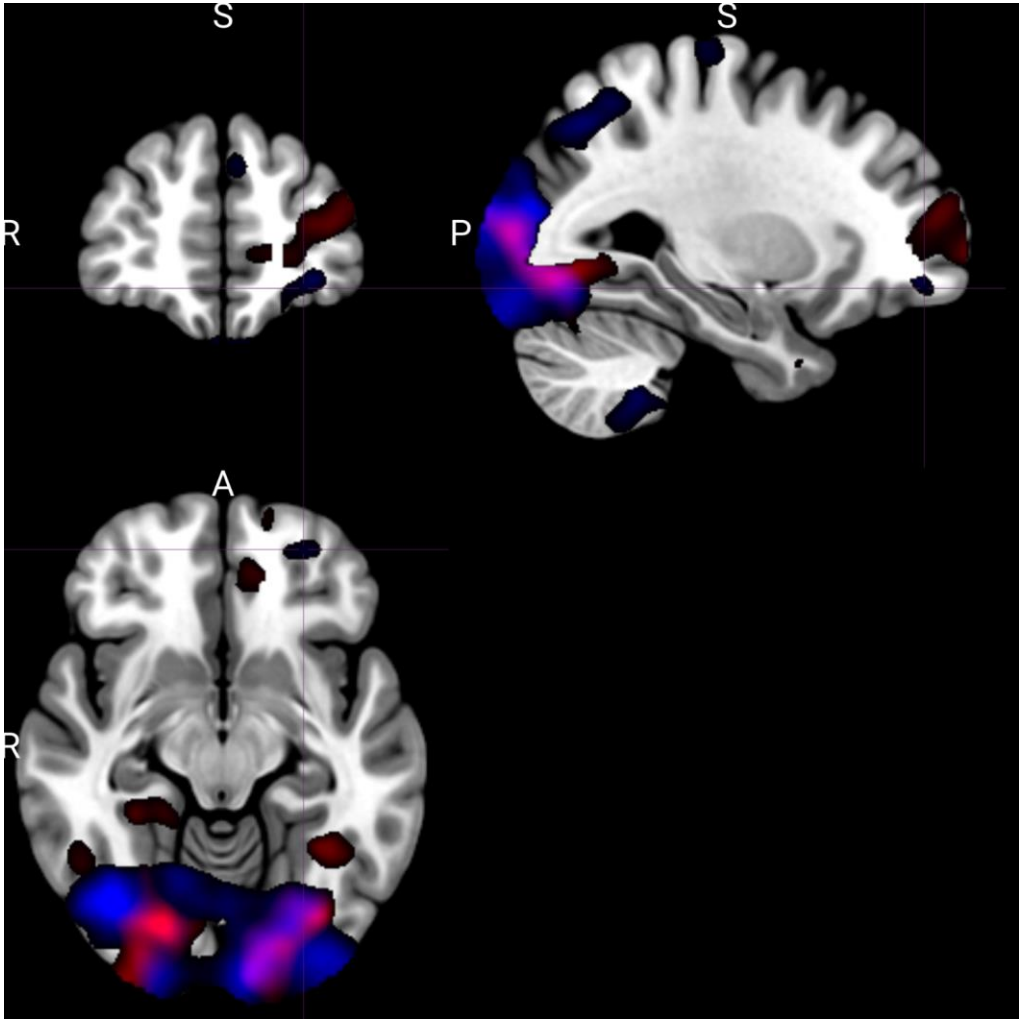
RSA on expected reward identity

Pre-meal data only

Odor-predictive cues (no clean air predictive cues)




Red: conditioned cues
Blue: preconditioned cues




Red: conditioned cues
Blue: preconditioned cues

# ROI-based analysis

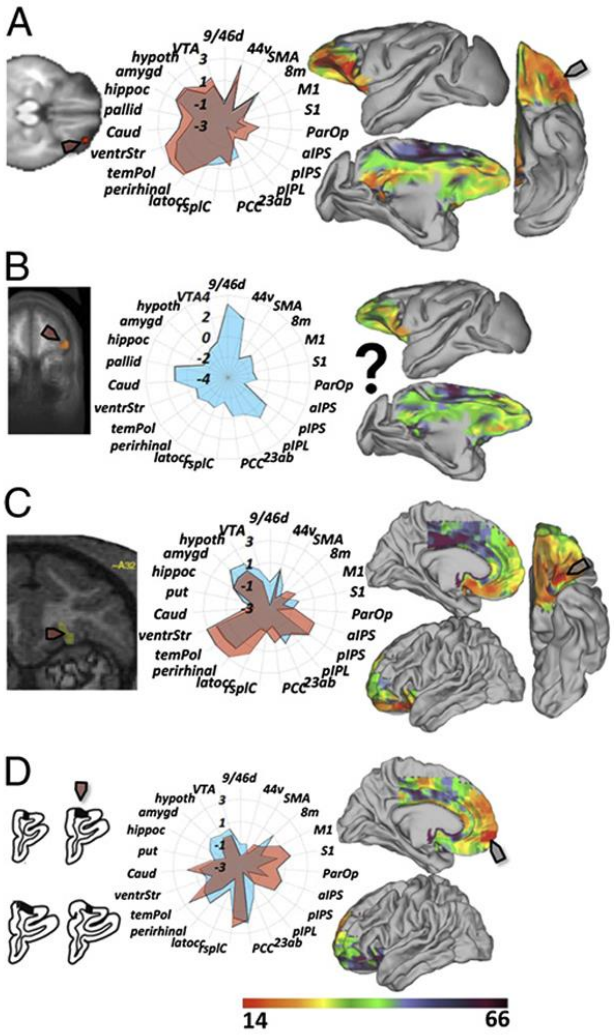
Neubert 2015 parcellation

Conditioned cues

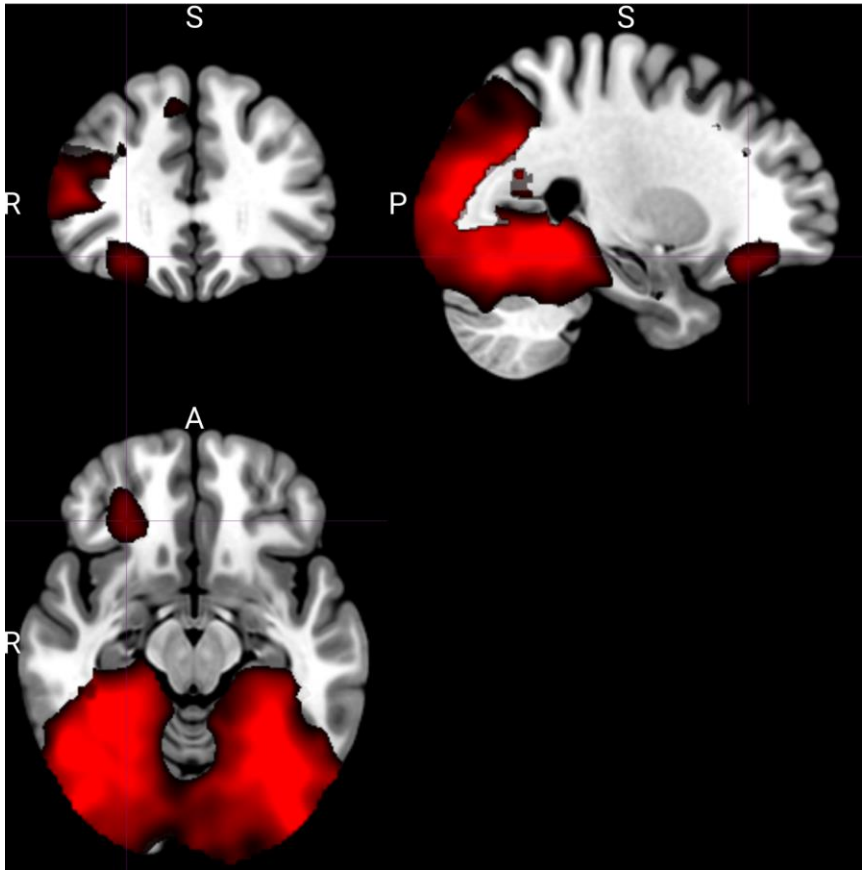
Left_14m, t(31) = 1.99, p = 0.0550

Preconditioned cues

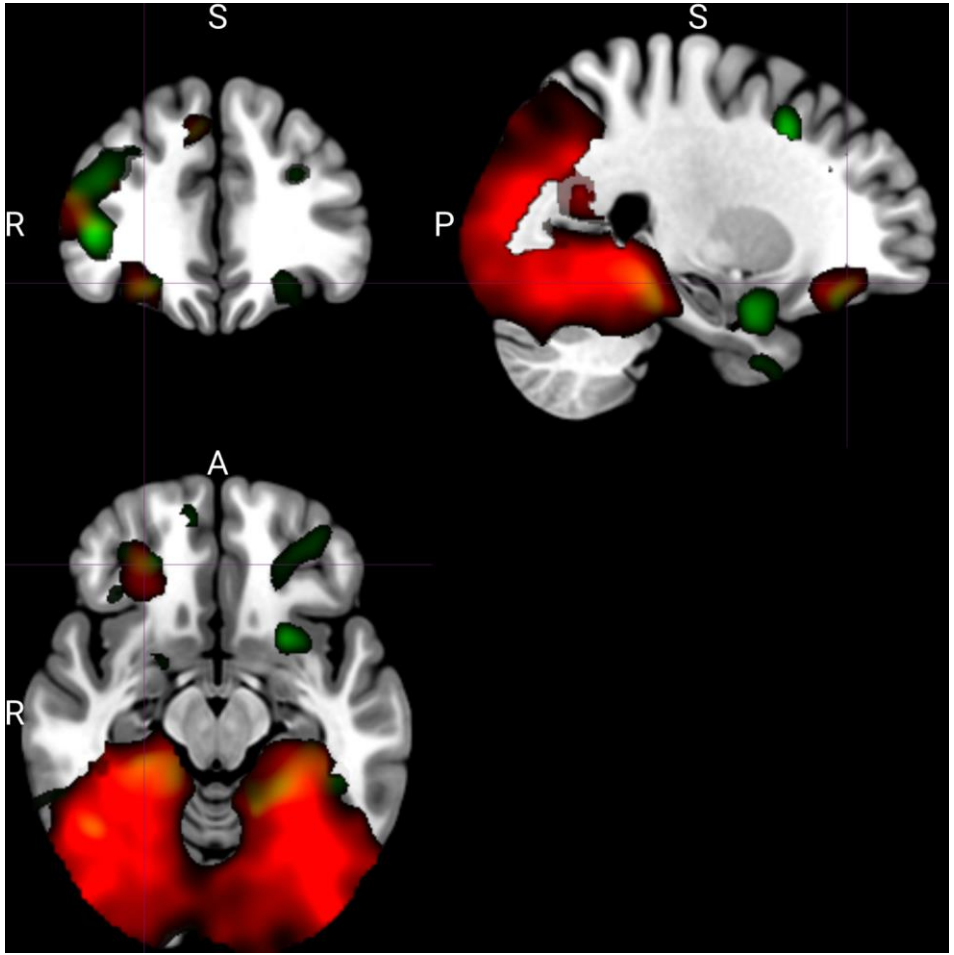
Right_11m, t(31) = 2.16, p = 0.0383



Different representation of cue-type (conditioned vs. preconditioned) identified from RSA



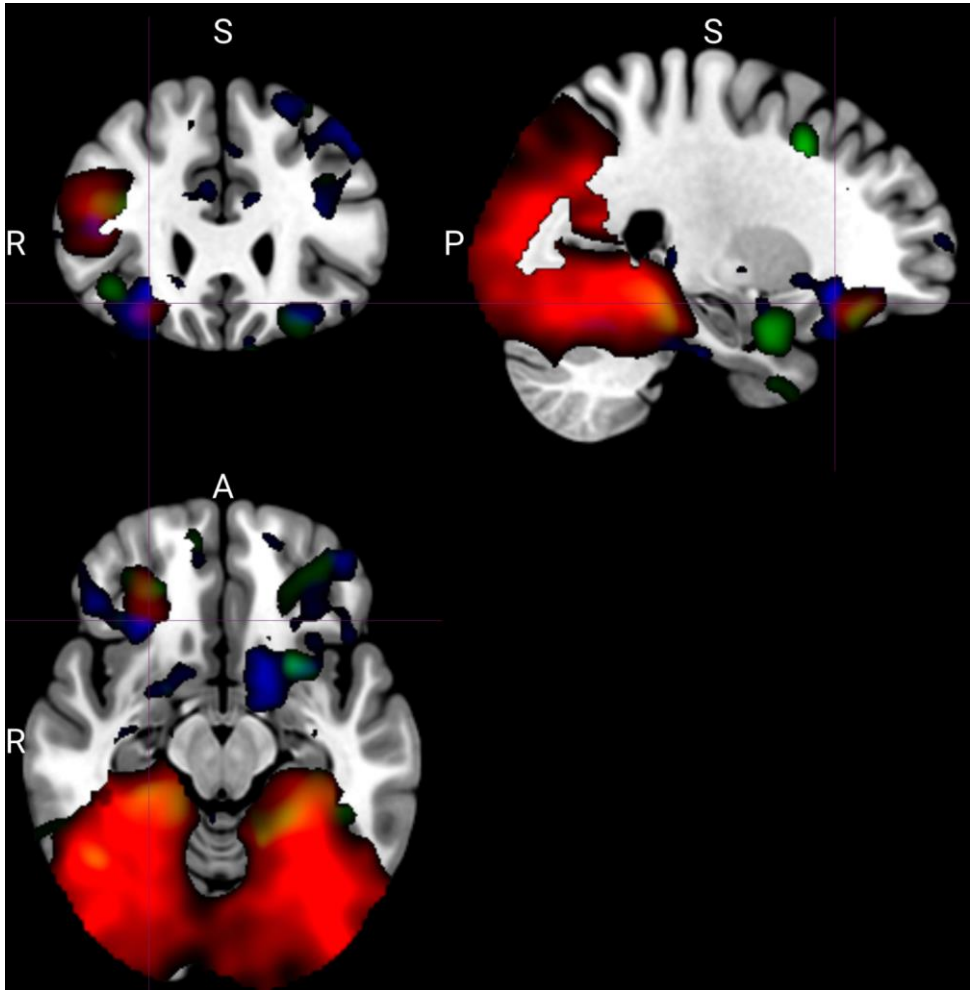
Decrease of such cue type representation from pre- to post-meal (green, thresholded >2, overlaid with red)



Univariate analysis:

Preconditioned > conditioned cues

(blue, thresholded >2)



Value RDM

- Associated odor pleasantness ratings as proxy for value
- Use data across pre and post meals

